

2.6 Identify data structures in problem scenarios

اگر ہر عنصر کا واضح اگلا/پچھلا ہے تو ڈھانچہ لکیری ہے؛ اگر شاخیں یا کئی روابط ہیں تو غیر لکیری۔

CS XII

Chapter 2: Computational Thinking & Algorithms

Lecture 22 of 24

2.6 Identify data structures in problem scenarios

LEARN IT

To pick a structure, first see how the data is organised. If each item has a clear "before and after", it is **linear**. If it branches or connects to many items, it is **non-linear**.

2.6.1 Linear scenarios

1. **Storing student grades for a class → Array.** You usually know how many students there are, and all grades are the same type. Indices give rapid lookup.
2. **Music playlist / browser history → Linked list.** Every song knows what comes next. When one finishes, follow the pointer. Frequent insert/delete is cheap.
3. **Undo or Redo → Stack (LIFO).** Each typing or formatting action is PUSHed. Undo POPs the most recent action.
4. **Printer task scheduling → Queue (FIFO).** Documents print in the exact order they were received. First-come, first-served.

2.6.2 Non-linear scenarios

1. **Organisational chart of a college → Tree.** Principal is the root, then deans, then HODs, then teachers. Clear parent-child links.
2. **GPS navigation and maps → Graph.** Cities are nodes, roads are edges. No strict top-down hierarchy. Several routes can reach the same destination.

If the data...	Then it is...	Recommended structure
Follows "next" or "previous" logic	Linear	Array, Linked list
Follows "parent" and "child" logic	Non-linear	Tree
Follows "first-come, first-served" logic	Linear	Queue
Follows "connections / relationships" logic	Non-linear	Graph
Needs to be accessed in the reverse order of entry	Linear	Stack

WORKED EXAMPLE

Worked example: match five school problems.

Scenario	Structure	Why (one line)
Store 40 practical marks you already counted	Array	Fixed size, same type, need A[i] lookup.
A playlist where the teacher inserts a nasheed between two songs	Linked list	Insert by moving two pointers, no big shift.
Ctrl+Z after formatting a heading	Stack	Last action must come off first.
Students lining up for the photocopier	Queue	First in line is served first.
Friend links in a class WhatsApp "everyone knows everyone"	Graph	Any student can connect to any other.

REMEMBER IT

Decision order: reverse-of-entry? → stack. First-come? → queue. Parent-child? → tree. Many-to-many routes? → graph. Known list with indexes? → array. Lots of insert/delete in a sequence? → linked list.

COMMON MISTAKE

Mistake: choosing an array for Undo because "actions are stored in a list".

Fix: the **access rule** matters more than "it is a list". Undo is reverse order of entry → stack.

اردو میں سمجھیں

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CHECK YOURSELF

1. Which structure stores a class's grades, and why?
2. Printer jobs: stack or queue? Why?
3. "Needs reverse order of entry" maps to which structure?

Answers: 1. Array — known size, same type, fast index lookup. 2. Queue, because first-come, first-served. 3. Stack.

Next lecture: 2.7 Trace data retrieval in Array, List and Queue